Clinton Enwerem

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Summary

Robotics Research Engineer and Electrical and Computer Engineering Ph.D. candidate at the University of Maryland, graduating in Fall 2026. Research in dexterous manipulation, robot learning, motion planning, and feedback control, with Python/C++ implementations in simulation and on physical arm–hand systems.

Skills

Robotics: ROS/ROS2, ros2_control, MoveIt2, MuJoCo, Drake, Isaac Sim, Isaac Lab, Pinocchio, SocketCAN; **Motion Planning:** RRT*, Informed RRT*, Sampling-Based Planning, FCL Collision Checking; **Control:** Control Barrier Functions, Safety Filters, Model Predictive Control (MPC), QP-based Control, Belief-Space Planning, Safe & Constrained Reinforcement Learning; **Robot Learning & Generative Models:** Imitation Learning & Behavior Cloning, ACT (Action Chunking Transformers), DAgger, Diffusion Policies (DP3), Flow-Matching Generative Models, Reinforcement Learning (PPO); **ML / Perception:** PyTorch, TensorFlow, OpenCV; **Programming:** Python, C++, MATLAB, Bash; **Optimization:** CasADi, CVXPY, qpsolvers, IPOPT, Gurobi; **Dev Tools:** git, Docker; **Robots & EoATs:** Fairino FR3, xArm7, Unitree G1 (simulation), LEAP Hand v1, RealHand L6.

Work Experience

Graduate Research Assistant — Aug. 2021 – Present
Institute for Systems Research (ISR), University of Maryland — College Park, MD

- Develop grasp-synthesis, motion-planning, and feedback-control methods for Fairino FR3 and xArm7 arms with LEAP Hand v1 and RealHand L6 end effectors.
- Build synchronized tactile, proprioceptive, and visual data pipelines; train imitation and reinforcement-learning policies; and evaluate learned manipulation in Isaac Lab.

Robotics Trainee — Jan. 2020 – Feb. 2021
Robotics & Artificial Intelligence Nigeria — Ibadan, Nigeria

- Built ROS-based visual-SLAM, state-estimation, and flight-control software for a low-cost UAV on embedded hardware.

Projects

ParcelStow, Expert–Learner Comparison Across Task Execution Speeds (Paper, Code, Data & Models) | *Isaac Lab, PyTorch, ACT, DAgger, Diffusion Policy* — 2026

- Built a Unitree G1 and RealHand L6 grasp–reorient–insert benchmark and trained three imitation-learning baselines on **297** expert demonstrations. The expert and ACT both reached **100%** nominal success; at 2× execution speed, success was **84%** for the expert and **53%** for ACT.

Grasp Distance Fields, Feedback Control for Grasp Execution (Project Page, arXiv) | *C++, Python, QP, CBF-CLF* — 2026

- Implemented a CBF-CLF quadratic-program controller over a configuration-space grasp distance field. Physical experiments lifted **46 of 50** objects, retained a **median 94%** of the certified grasp margin, and solved the QP in **0.09 ms**.

`realhand_l6_ros2`, Dexterous-Hand ROS 2 Stack (Code, DOI) | *C++, ROS 2, ros2_control, SocketCAN* — 2026

- Implemented a `hardware_interface::SystemInterface`, tactile-state export, contact-gated control, URDF/SRDF, MoveIt 2 configuration, mock hardware, virtual-CAN tests, and CI. Validated CAN communication, joint feedback, and tactile acquisition on a physical RealHand L6.

Selected Publications

- **C. Enwerem**, J. S. Baras, C. Belta. "Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert–Learner Comparison Across Task Execution Speeds." *arXiv preprint*, 2026. arXiv:2609.01453, code, data & models.
- **C. Enwerem**, J. S. Baras, C. Belta. "Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality." *Preprint*, 2026. arXiv:2608.00600, project page.
- Also *to appear in 2026* at IROS and CDC, with five additional peer-reviewed first-author papers in control, safe learning, and robotic manipulation.

Education

Ph.D., Electrical & Computer Engineering, University of Maryland, College Park, MD. Expected Fall 2026. — Aug. 2021 – Present
B.Eng., Electrical Engineering, University of Nigeria, Nsukka. Highest Honors. GPA 3.84/4.0. — Aug. 2018